

CONSTRUCTION DATA	
DESIGN CODE	ASME SECTION VIII DIV 1 - EDITION 2015
ASME U-STAMP	NOT REQUIRED
DESIGN PRESSURE (MAX.) INTERNAL	11.4 Mpa(g) [1654 Psi(g)] [114 bar(g)]
DESIGN TEMPERATURE INTERNAL (MAX)	100 °C [212 °F]
MINIMUM DESIGN METAL TEMPERATURE	0 °C [32 °F]
CORROSION ALLOWANCE (INTERNAL)	1.5 mm [0.059 inch]
HYDROSTATIC TEST PRESSURE (SEE NOTE 2)	17.1 Mpa(g) [2480 Psi(g)] [171 bar(g)]
EMPTY WEIGHT	530 kg [1168 lbs]
SURFACE PREPARATION AND PAINT	SEE NOTE 3

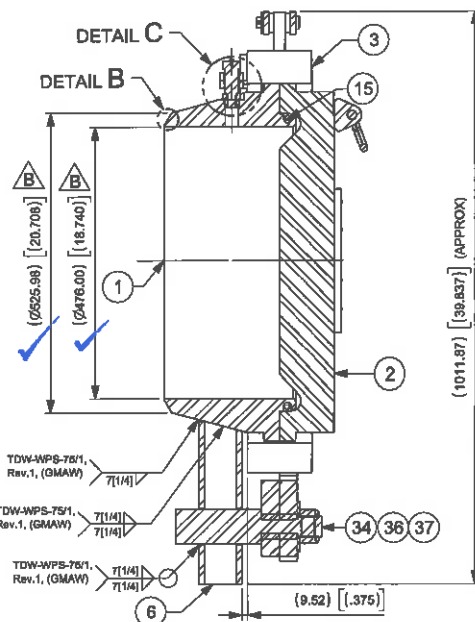
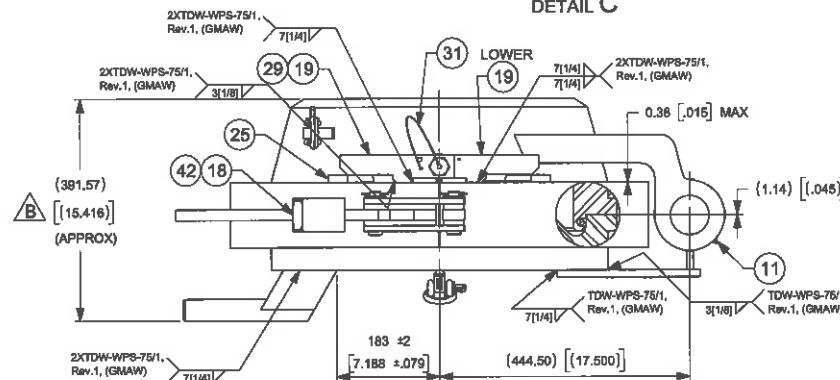
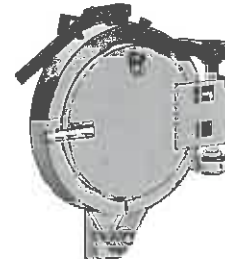
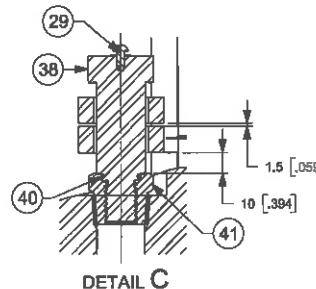
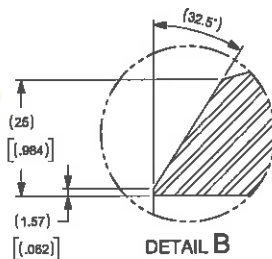
NOTES:

- [1] ALL DIMENSIONS ARE IN mm[inch] UNLESS OTHERWISE SPECIFIED.
- [2] TDW SHOP HYDROSTATIC TEST IS NOT REQUIRED.
- [3] SURFACE PREPARATION & PAINTING:
SURFACE PREPARATION: SHOT BLASTING TO SA 2 1/2
PAINT: WILLIAMSON RED
DFT: 25 TO 30 MICRON
PROTECT ALL THREADS, NAME PLATES, O-RING GROOVE & BEARINGS FROM PAINT & SHOT BLASTING
- [4] MANUFACTURING CANNOT BEGIN UNTIL A PROPERLY SIGNED AND DATED DRAWING CHECKED "RELEASED FOR FABRICATION" OR "RELEASED FOR FABRICATION EXCEPT AS NOTED" IS RETURNED.

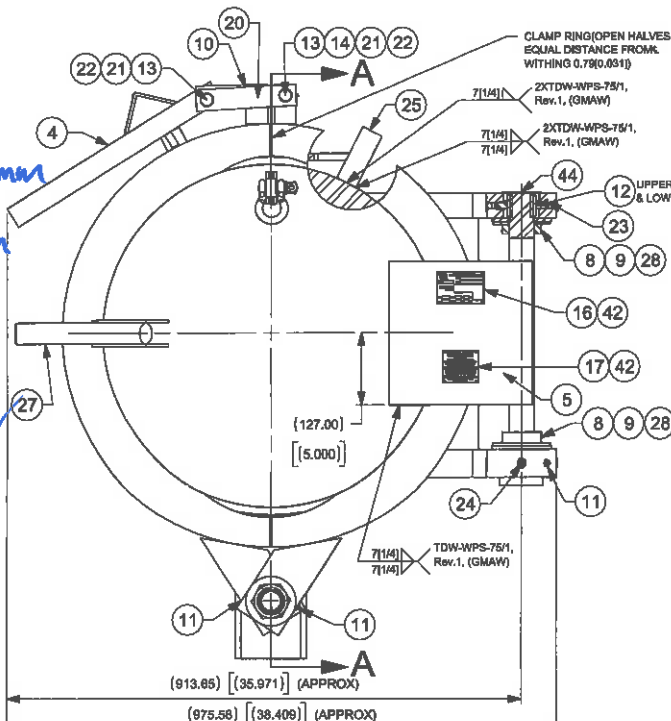
FOR CUSTOMER APPROVAL

- ☒ RELEASED FOR FABRICATION
☐ RELEASED FOR FABRICATION EXCEPT AS NOTED
☐ NOT RELEASED FOR FABRICATION CORRECT AND RESUBMIT

CUST. P.O. NO.: **2166/17-006**
19/7
 CUSTOMER SIGNATURE DATE



Handwritten notes:
 PIPE CD = 508 mm
 PIPE WT = 16 mm
 PIPE ID = 508 - 32 = 476 mm



45	00-3795-0610-00	OPERATION & MAINTENANCE MANUAL (NOT SHOWN)	1
44	00-5581-0002-00	CAPLUG	1
43	00-3795-0283-00	INSTRUCTION (NOT SHOWN)	1
42	00-2905-0300-03	DRIVE SCREW	12
41	00-5213-0014-00	PRESSURE WARNING SCREW BUSHING	1
40	00-1245-0007-00	O-RING, .0625in	FKM 1
39			
38	00-4576-0037-00	PRESSURE WARNING LOCK SCREW	1
37	00-0231-0014-00	ESNA NUT, TYPE T6 SERIES HEAVY THIN	1
36	00-0001-0015-00	WASHER, 1.625in ID	1
35			
34	00-5211-0002-05	CLAMP RING PIVOT PIN	1
33			
32			
31	00-4366-0003-00	WIRE ROPE LANYARD	1
30			
29	00-2905-0700-08	DRIVE SCREW	2
28	00-5574-0004-00	RUBBER SEAL	2
27	00-4787-0001-01	DOOR HANDLE	1
26			
25	00-4886-0073-01	CLAMP RING, STOP	2
24	00-4002-0005-00	CAPLUG	8
23	00-5505-0800-07	SET SCREW, 0.500in - 13X0.875in	8
22	00-5509-0400-20	COTTER PIN, 0.125inX1.50in LONG	2
21	00-0435-0010-00	WASHER, SAE 0.750in	2
20	00-4228-4824-14	SPIROL PIN	1
19	18-3724-0619-00	CLAMP RING LOCKING LUG	2
18	00-3081-0043-00	PRESSURE WARNING PLATE	1
17	12320461	PRESSURE WARNING PLATE	1
16	12320454	NAME PLATE	SS304/SS302 1
15	19-3700-1925-53	O-RING, .500in THK X 19.25in ID	FKM 2
14	00-4011-0008-00	SPACER	1
13	00-1179-0007-00	CLEVIS PIN	2
12	00-5214-0001-00	BEARING ASSEMBLY	2
11	00-0112-0001-00	GREASE FITTING	4
10	00-4613-0048-00	HANDLE LINK	2
9	00-0001-0017-00	FLAT WASHER, 1.75in	2
8	00-6683-0006-12	COLLAR, SHAFT Ø 1.75in	2
7			
6	12336878	CLAMP RING PIVOT PLATE, 20in	1
5	12336850	DOOR HINGE ARM, D2000 20in	1
4	00-4798-0002-01	CLAMP RING HANDLE	1
3	12336848	CLAMP RING, D2000 20in	SA105 1
2	12336846	DOOR, D2000 20in SECTION VIII	SA105 1
1	12336844	BARREL COLLAR, D2000 20in RH SECTION VIII	SA105 1

BILL OF MATERIAL						
E						
D						
C						
B	REVISED AS PER COMMENTS	12JUL17	MJV	PV	KS	PV
A	NEW RELEASE	08JUL17	MJV	PV	KS	PV
REV	DESCRIPTION	DATE	DRN	CHK	WELD ENGR	APRD
Dimensions and Tolerances per ASME Y14.5		TOLERANCES AND THE DESIGNER IS THE OWNER OF IT. WILLIAMSON INDIA PRIVATE LIMITEDS AND MUST NOT BE COPIED OR REPRODUCED WITHOUT THEIR PERMISSION IN WRITING		 T D Williamson India Private Limited SAVLI, VADODARA		
MACHINING TOLERANCES		DRAWN BY	MJV	TITLE		
UNLESS OTHERWISE SPECIFIED TOLERANCES PER TITLE BLOCK APPLY. FINISH ALL MACHINED SURFACES. BREAK ALL SHARP CORNERS TO 0.06mm RADIUS MAX. MINIMUM TOLERANCE: 0.1mm. ANGULAR TOLERANCE: ±10°		CHECKED BY	PV	CLOSURE, D2000 20in 0.984in RH SECTION VIII		
		WELD ENGR	KS	SIZE	DRG. NO.	
		APPROVED BY	PV	A3	12336843	
		DATE	06JUL17	SCALE: NTS	SHEET 1 OF 1	REV B



T.D. Williamson Inc.

D2000 Closure Calculation
ASME Section VIII-1, 2015 Edition

Ames
19/7/17
PONO: 2166/17-006

PN: 12336843 TDW Document No:-12336843-CALC Rev.B

Size: 20in CLOSURE

Design Pressure..... 1654 PSI
Design Temperature..... 212 °F
Corrosion Allowance.... 0.059 in
MDMT..... 32 °F

Barrel Collar Material..... ASME SA105N
Door Material..... ASME SA105N
Barrel Allowable Stress (MASV).... 20000 psi
Door Allowable Stress (MASV)..... 20000 psi

Closure Shell (Hub) ASME Section VIII-1 para UG-27(c):

Radius of Hub..... R = 9.429 in
Joint Efficiency..... E = 1.000
Barrel Allowable Stress. S = 20000 psi

$$t = \frac{PR}{SE - 6P} + CA = 0.880 \text{ in Minimum}$$

Design Thickness, t_D = 0.984 in

Closure Head (Door) ASME VIII-1, UG 34(c)(2):

Short Span/Diameter.... d = 20.697 in
Construction Factor..... C = 0.30
Joint Efficiency..... E = 1.000
Door Allowable Stress.. S = 20000 psi

$$t = d \sqrt{\frac{CP}{SE}} + CA = 3.319 \text{ in Minimum}$$

Design Thickness, t_H = 3.8 in

PRINTED ON: 12-Jul-17

LATEST REVISION:

C



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NOTES:

BY/DATE:

MJV

12-Jul-17